

Continuous-Time Delta-Sigma Modulators With Time-Interleaved FIR Feedback

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Abstract—The use of a single-bit quantizer in a wideband CT $\Delta\Sigma$ is attractive, as the quantizer can be implemented in a power and area-efficient manner. Unfortunately, 1-bit CT $\Delta\Sigma$ s are plagued by a host of difficulties. Clock jitter and quantizer metastability are particularly problematic, and the higher loop filter linearity needed to process the full-scale feedback waveform results in increased power dissipation. The use of a finite impulse response (FIR) feedback DAC is a power efficient way of addressing the challenges above. However, even this technique runs into difficulties at multi-GHz clock rates. This paper introduces the idea of time-interleaved (TI) FIR feedback to enhance the performance of a conventional FIR DAC. A single-bit CT $\Delta\Sigma$ that uses a $2\times$ TI-FIR DAC achieves 67.6/68.8/76 dB SNDR/SNR/DR in a 60 MHz bandwidth. Designed in a low leakage 65 nm CMOS process, the modulator operates at 6 GHz and occupies only 0.07 mm². Its Walden Figure of Merit is 56.5 fJ/1v1.

Index Terms—Oversampling, single-bit, FIR DAC, jitter, distortion.

I. INTRODUCTION

ANALOG-TO-DIGITAL converters with signal bandwidths in the 50 MHz range are necessary in several emerging communication applications. Continuous-time $\Delta\Sigma$ modulators (CT $\Delta\Sigma$ s) are attractive low power alternatives to Nyquist ADCs in such signal chains due to the ease with which they can be driven, their inherent anti-aliasing, and the relaxed speed requirements on the active elements. Several CT $\Delta\Sigma$ s targeting signal bandwidths of around 50 MHz, realized in advanced nanometer CMOS technologies, and achieving 11–14 bit resolution have been reported recently [1]–[5]. All of these are multibit designs, which exhibit low sensitivity to clock jitter and quantizer metastability. Further, multibit operation also relaxes the linearity requirements of the loop filter. On the other hand, the quantizer (and the associated clock distribution circuitry) dissipates a significant fraction of the total power consumed by the modulator. Further, using dynamic element matching (DEM) to linearize the feedback DAC is difficult at the multi-GHz clock rates needed to achieve a high signal bandwidth. This has prompted many designs to resort to calibration, and/or use higher supply voltages to

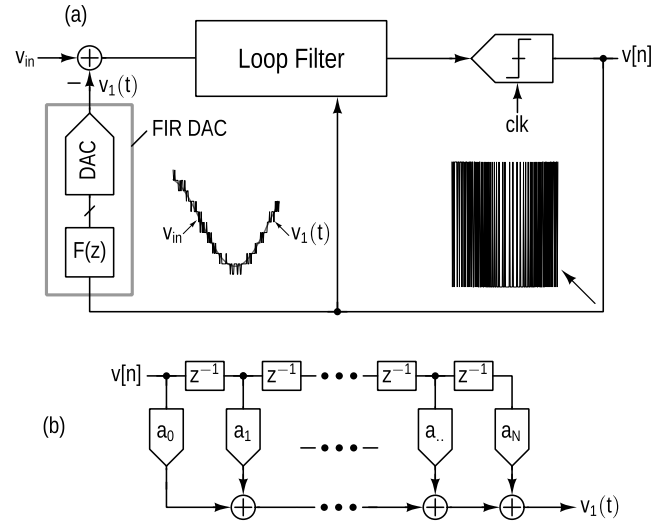


Fig. 1. (a) Operating principle of a 1-bit CT $\Delta\Sigma$ with an FIR Feedback DAC and (b) Semidigital implementation of the FIR DAC.

realize DACs with inherently low mismatch. Thus, in spite of the attractive aspects of the multibit architecture, it is challenging to implement at high speeds.

A single-bit CT $\Delta\Sigma$ is attractive due to its simplified ADC, which results in reduced power dissipation. Since the feedback DAC is inherently linear, DEM is not needed. However, the full-scale feedback waveform renders the CT $\Delta\Sigma$ unduly sensitive to clock jitter, and places increased demands on the linearity of the loop filter. Further, 1-bit operation necessitates a higher sampling frequency needed to achieve the desired signal-to-quantization-noise ratio SQNR, which reduces the regeneration time available for the latches in the ADC. The resulting data-dependent delay of the quantizer results in an increased in-band noise floor, much like the effect of clock jitter. It is thus seen that single-bit operation greatly simplifies the implementation of the quantizer, but causes a host of other difficulties.

The problems with the 1-bit quantizer can be addressed by the use of an FIR feedback DAC. The basic idea behind this approach is shown in Fig. 1(a). The 1-bit output of the CT $\Delta\Sigma$ is filtered by an FIR filter with transfer function $F(z)$ before being fed back. Thanks to FIR filtering, the step-size of the feedback waveform ($v_1(t)$) is reduced. This benefits the CT $\Delta\Sigma$ in two ways: apart from lowering jitter sensitivity, the modulator's linearity is improved since the loop filter now processes a much smaller signal. Further, the chain of flip-flops in the FIR DAC reduce the effect of latch metastability. If the

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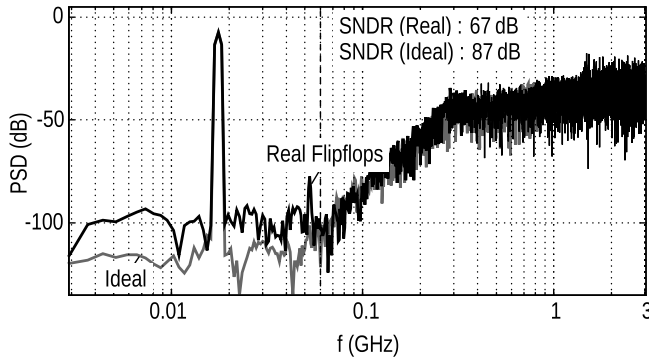


Fig. 2. Simulated power spectral density (PSD) of a fourth order single-bit CT $\Delta\Sigma$ M with an 8-tap FIR feedback DAC, operating at 6GS/s, with ideal and real flip-flops, in a 65 nm low-leakage CMOS technology.

FIR DAC is implemented in a semi-digital fashion, as shown in Fig. 1(b), element mismatch does not cause nonlinearity; it only modifies the frequency response of the FIR filter, which is benign. A single-bit quantizer with FIR feedback, therefore, combines the benefits of single-bit and multi-bit operation. [6], [7] are examples of wide-band CT $\Delta\Sigma$ Ms that owe their power efficiency and performance to FIR feedback.

What are the factors that limit the speed of operation (and thereby the signal bandwidth) of a single-bit CT $\Delta\Sigma$ M that uses FIR feedback? The delay of the fast path around the quantizer imposes a fundamental limit on the speed at which the modulator can be clocked. It turns out, that in spite of using FIR feedback, quantizer metastability can severely degrade in-band noise of the modulator at speeds much lower than this fundamental limit. To give a context to this, consider the CT $\Delta\Sigma$ M reported in this work, which attempts to achieve 12 bit resolution in a 60 MHz bandwidth in a low leakage 65nm CMOS technology. Analysis shows that a fourth order modulator operating with an oversampling ratio (OSR) of 50 is needed to achieve the desired SQNR of about 90dB. As mentioned earlier, at the 6GHz clock rate needed, even the FIR DAC approach begins to run into difficulties due to the limited time available for regeneration. This results in an appreciable data dependent jitter in the feedback waveform and manifests as an increased in-band noise floor, as demonstrated in Fig. 2. The figure shows the simulated power spectral density (PSD) of the 1-bit CT $\Delta\Sigma$ M (using transistor level latches and flip-flops) which employs an 8-tap FIR DAC, operating at 6GHz in a 65nm CMOS process. The ideal PSD, which has an SQNR of about 87dB in a 60MHz bandwidth for a -6dBFS input, is also shown for comparison. It is seen that the in-band SQNR is degraded by around 20dB as a consequence of data dependent jitter.

This work introduces the use of a time-interleaved (TI) 1-bit ADC and FIR DAC to address the speed limitations of a conventional ADC+FIR-DAC design. An immediately apparent benefit of interleaving is the significantly increased regeneration time for the comparators, which greatly reduces data dependent jitter in the feedback waveform. It turns out that time-interleaving also results in reduced power dissipation of the FIR-DAC. These aspects and other design considerations for CT $\Delta\Sigma$ Ms with time-interleaved FIR feedback form the bulk of the rest of the paper. To complement the discussion,

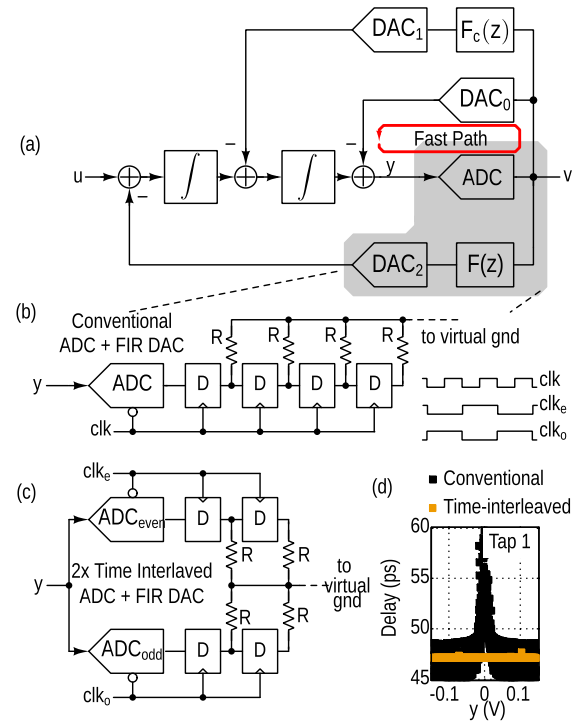


Fig. 3. (a) Second order CT $\Delta\Sigma$ M with an FIR DAC, chosen for illustration. (b) Conventional implementation of an FIR DAC with all equal taps. (c) Time-interleaved FIR DAC implementation. (d) Simulated data-dependent delay at the output of Tap-1.

design details and experimental results from a CT $\Delta\Sigma$ M test chip are given. This work is an extended version of [8], and is organized as follows. Section II gives the basic idea, describes the potential benefits of time-interleaving, analyzes various error mechanisms and the corresponding mitigation techniques in CT $\Delta\Sigma$ Ms with TI-FIR DACs. In particular, we show that time-interleaving can reduce the power dissipation of the FIR DAC by a factor of ≈ 2 . Section III gives the circuit design details of a fourth order CT $\Delta\Sigma$ M that uses a $2\times$ TI-FIR DAC to achieve 12-bit resolution in a 60 MHz bandwidth. Measurement results, presented in Section IV, show that the CT $\Delta\Sigma$ M that achieves SNR/SNDR/DR of 68.8/67.6/76 dB in a 60 MHz BW while sampling at 6GS/s. Implemented in a 65nm-LP process, it consumes only 13.3 mW and occupies 0.07 mm^2 , resulting in a FoM of 56.5 fJ/1v1. Section V gives the conclusions.

II. TIME INTERLEAVED FIR FEEDBACK

The idea behind time-interleaved FIR feedback is explained using the example second-order modulator of Fig. 3(a). For purposes of illustration, we assume that a 4-tap FIR DAC is used, and that DAC_2 is implemented using resistors. The resistors feed into the virtual ground of an active-RC integrator in the loop filter. DAC_0 is the direct-path DAC, while DAC_1 and $F_c(z)$ represent the compensation FIR DAC. In a full-rate modulator, the ADC- DAC_2 - $F(z)$ combination would be implemented as shown in Fig. 3(b). The ADC is clocked on the falling edge of the clock, while the flip-flops that comprise the DAC are clocked on its rising edge. This way, an explicit half clock cycle delay is introduced into the loop, with the idea that the (single-bit) ADC has enough time to regenerate, thereby

greatly reducing data-dependent jitter in the output waveform of the main FIR DAC. The compensation FIR DAC ($F_c(z)$ and DAC_1) and the direct-path DAC (DAC_0) are needed to stabilize the loop, which would otherwise be rendered unstable by the FIR DAC and the deliberately introduced half cycle delay.

The compensation FIR DAC and the direct path DAC waveforms will also be afflicted by data-dependent jitter; however, the effect of this on the in-band SNR will be small, due to the high gain of the input integrator in the signal band. Thus, as far as data-dependent jitter is concerned, the outermost DAC ($F(z)$ - DAC_2) is of the biggest concern. At very high clock speeds, like the 6GHz rate used in this work, even half-cycle delay between the ADC and DAC clocks does not adequately suppress data-dependent jitter. As a result, the output of the first tap of the FIR DAC has significant data-dependent jitter. This is shown in Fig. 3(d), from which it is seen that the delay at the output of the first tap can increase by as much as 15ps from the nominal delay, when the output of the loop filter is small. As a result, the in-band noise floor is raised. The time interleaved ADC and FIR DAC aim to address this fundamental problem of inadequate regeneration time, as discussed below.

Fig. 3(c) shows the ADC and FIR DAC combination implemented in a time-interleaved fashion. The full rate clock is used to generate two half-rate clock signals, termed clk_o and clk_e , spaced 180° apart. (The subscripts e and o are intended to denote even and odd.) clk_e and clk_o drive the even and odd ADC + FIR DAC channels respectively. Since the intention is to realize the equivalent of a 4-tap full-rate FIR DAC, the time interleaved sections need to have two taps each. In each section, the ADC is clocked on the falling edge, while the flip-flops are clocked on the rising edge. Thanks to the half-rate clocks, the time available for regeneration is doubled when compared to the conventional ADC and FIR DAC, as shown in Fig. 3(b). As a result, data dependent jitter at the output of the first tap is drastically reduced, as placed in evidence by Fig. 3(d). Thus, the in-band SQNR of a (properly stabilized) CT Δ ΣM that uses a time interleaved FIR DAC is not limited by data-dependent jitter at the output of its flip-flops.

Referring to Fig. 3(a), the fast path around the quantizer (through DAC_0) needs to be closed at the full-rate of the modulator. This represents the critical path. To maintain the same noise transfer function (NTF) with a time-interleaved ADC and FIR DAC, the fast loop must still be closed within one period of the full-rate clock. This can be accomplished by driving DAC_0 with the multiplexed outputs of ADC_{odd} and ADC_{even} . Since the fast path is a lot more tolerant of data-dependent jitter, one need not wait for very long to “look” at ADC_{odd} and ADC_{even} for a valid output; our design waits for about half-clock cycle of the full-rate clock. Adding to this the multiplexer delay of about quarter-cycle, we see that the critical path can still be closed within one cycle of the full-rate clock.

The key point to note from the discussion above is that time-interleaving aims to eliminate the increase in in-band noise due to metastability in the outer-most DAC of the modulator. The

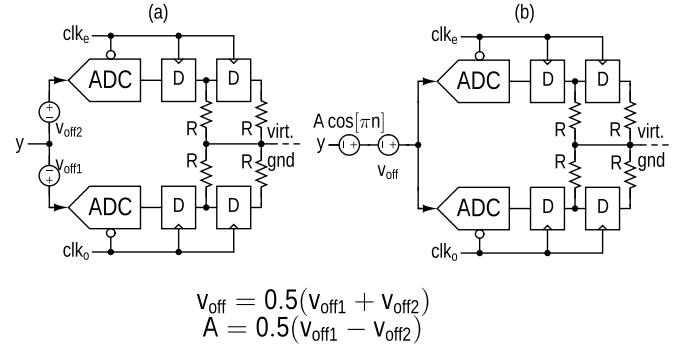


Fig. 4. Modeling the effect of DC offset mismatch in the interleaved sections of the ADC. (a) Time-interleaved ADC with offsets. (b) Equivalent model.

fast loop must still be closed within one full-rate clock cycle—this path can accommodate a lot of data-dependent jitter, since jitter noise due to the innermost path, when referred to the modulator input is greatly reduced by the large in-band gain of the loop filter.

Another happy consequence of time interleaving the DAC is the reduction in its dynamic power dissipation. As seen from Fig. 3(b) and (c), the conventional FIR DAC uses 4 flip-flops clocked at the sampling rate, while its interleaved equivalent employs 4 flip-flops operating at half the rate. Assuming CMOS logic, this means that the power dissipated in the flip-flops of the time-interleaved FIR DAC is smaller by a factor of two. The observation above begs the question – “why not realize the FIR DAC using four interleaved parts, each operating at quarter the rate? This should reduce the power dissipation and reduce data-dependent jitter further, isn’t it?” While this is true in principle, there are practical problems associated with this approach. First, there is the difficulty of generating and distributing multiple low-frequency clocks. More importantly, like in any time-interleaved system, mismatch between the interleaved sections degrades modulator performance. Mismatch between channels can be attributed to several factors – offset mismatch in the ADCs, timing errors of the clocks exciting each interleaved section, and mismatch between the FIR filters in each channel. We analyze these factors next.

A. Offset Mismatch

The single-bit ADCs in each individual section can have different dc offsets, denoted by v_{off1} and v_{off2} , as shown in Fig. 4(a). It is easy to see that this is equivalent to adding a dc offset of $0.5(v_{off1} + v_{off2})$ and a tone (at $f_s/2$) of amplitude $0.5(v_{off1} - v_{off2})$ to the output of the loop filter in an offset-free CT Δ ΣM, as shown in Fig. 4(b). The dc offset is suppressed by the loop’s NTF, while the tone at half the sampling frequency lies outside the signal band and will be eliminated by the decimation filter. It is thus seen that offset between comparators is benign.

B. Timing Skew

Next, we consider timing skew. Ideally, clk_e and clk_o are exactly T_s apart. In practice, there will be timing skew between these clocks. For purposes of analysis, we assume that clk_o is skewed as shown in Fig. 5(b). The current provided by the

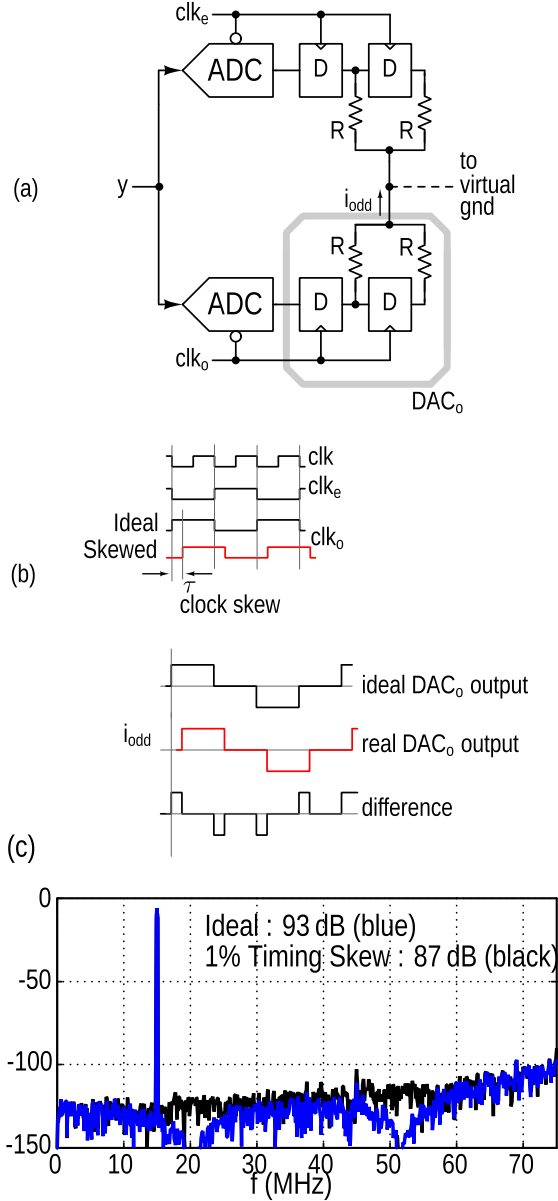


Fig. 5. (a) Modeling the effect of timing mismatch in the interleaved sections of the DAC. (b) Ideal and real clock and DAC output waveforms. (c) In-band PSD with and without clock skew.

“odd” DAC (i_{odd}) is skewed by τ with respect to the ideal waveform, as shown in the figure. The error current waveform can be interpreted as processing the ideal waveform through a filter with transfer function $(1 - e^{-s\tau})$. Since the PSD of the error current is first-order shaped, the effect of timing skew on the in-band signal to noise and distortion ratio (SNDR) can be expected to be small. This is confirmed by the simulation results shown in Fig. 5(c), which shows the in-band spectra for the fourth order CTD $\Sigma\Delta$ of the paper employing an 8-tap FIR DAC implemented using $2\times$ time-interleaving. The peak SNDR, which is ideally about 93 dB, is only degraded by 6 dB (in a 60 MHz bandwidth) for a skew of 1.7 ps, which is 1% of the period of the full rate clock.

Fig. 6 shows the simulated in-band SNDR as a function of clock skew. The targeted thermal-noise limited SNDR of our design is about 75 dB. Monte-carlo simulations on the clock-generator circuitry show a worst-case skew between the

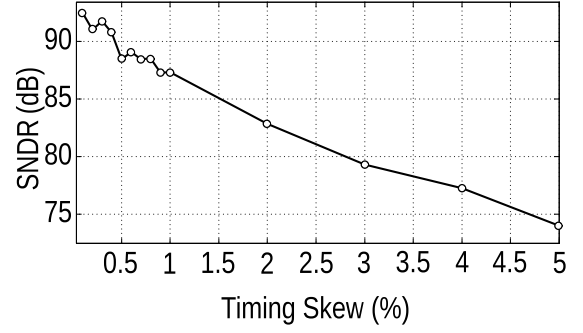


Fig. 6. Simulated in-band SNDR as a function of clock skew.

interleaved clocks of about 3.5 ps. This will limit the SNDR of our modulator to about 81 dB.

C. FIR Filter Mismatch

The third, and most important, manifestation of mismatch is that due to resistors used in the FIR feedback DACs. Resistor mismatch causes differences between the transfer functions of the FIR DACs in each interleaved section. Fortunately however, it is only the mismatch in the DC gains of the interleaved DACs that is relevant, as we show next. The analysis is somewhat involved, and the patience of the reader is greatly appreciated. To aid understanding, we use the conceptual signal processing model of the time-interleaved FIR DAC shown in Fig. 7(a). $v[n]$ represents the full-rate output sequence of the modulator. The output waveforms of the even and odd DACs, denoted by $v_e(t)$ and $v_o(t)$, can be thought of as being produced as follows.

We first form the $2\times$ interpolated even and odd sequences, denoted by $v_{e2}[n]$ and $v_{o2}[n]$ as follows.

$$\begin{aligned} v_{e2}[n] &= 0.5 \cdot (1 + \cos[\pi n]) \cdot v[n] \\ v_{o2}[n] &= 0.5 \cdot (1 - \cos[\pi n]) \cdot v[n] \end{aligned} \quad (1)$$

v_{e2} and v_{o2} excite nominally identical FIR filters with transfer function $F(z)$, whose outputs drive the even and odd NRZ DACs, denoted by DAC_e and DAC_o respectively. The DAC pulse width is T_s . Recall that, in the time-interleaved FIR DAC, the outputs of the even and odd DACs are held for two bit periods, corresponding to a time duration of $2T_s$. In the model of Fig. 7(a), this is enforced by constraining successive taps of the FIR filters to be equal. Thus, the first and second tap weights of the filter are equal (and denoted by f_0), the third and fourth taps are equal etc. As a result, the transitions on $v_e(t)$ occur only at even multiples of n , while those on $v_o(t)$ occur at odd multiples of n .

Since DAC_e and DAC_o are identical, the diagram of Fig. 7(a) can be conceptually simplified, as shown in Fig. 7(b), where the outputs of the FIR filters are first added to yield the full-rate sequence $v_f[n]$, before exciting the DAC. It is clear that

$$V_f(z) = (V_{e2}(z) + V_{o2}(z))F(z). \quad (2)$$

From (1), $V_{e2}(z) + V_{o2}(z) = V(z)$, so it follows that $V_f(z) = F(z) \cdot V(z)$, indicating that the schematics of Figs. 7(a) and (b) are equivalent to a full-rate FIR DAC with transfer function $F(z)$. The constraint imposed by time-interleaving is that

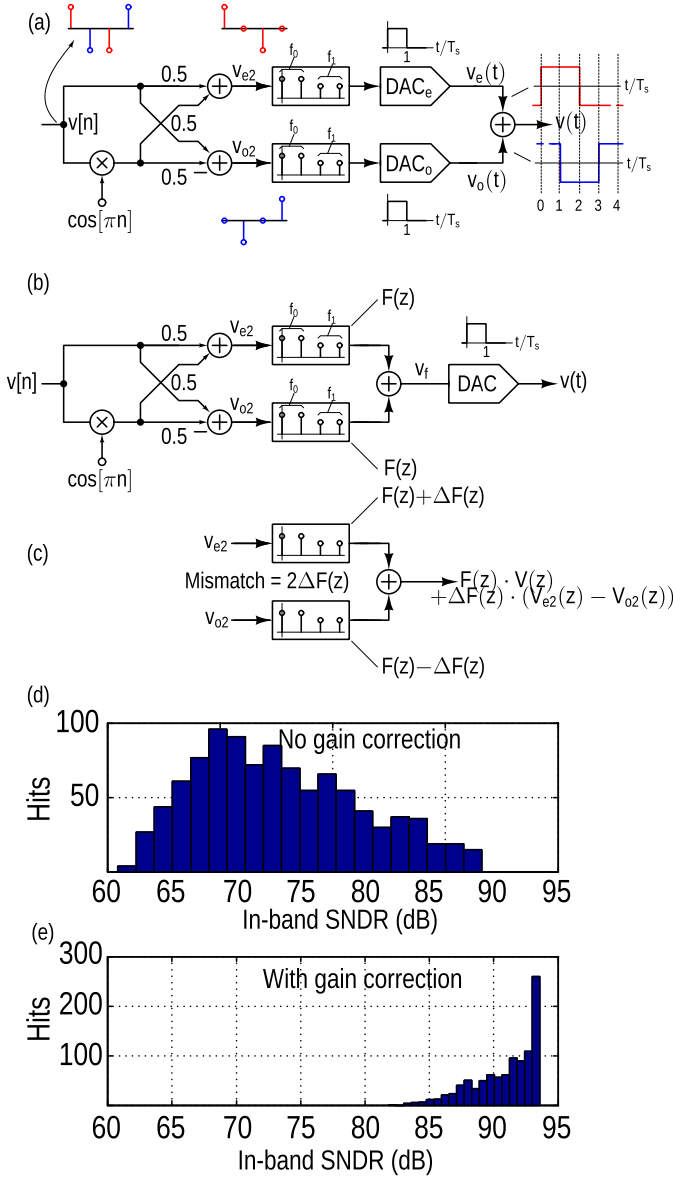


Fig. 7. (a) Signal processing model of the time-interleaved FIR DAC. The NRZ DACs operate at the full rate. (b) Equivalent model. (c) Model with mismatch in the filter transfer functions. (d) Monte Carlo simulation of a CTΔΣM with mismatched weights of the 8-tap FIR-DAC, and (e) after calibrating for only the dc gain of the DACs.

successive tap weights of the FIR filter are equal. It turns out that this is not overly restrictive in practice.

If the FIR filters are mismatched (due to mismatch in the resistor values), their transfer functions can be expressed as $F(z) + \Delta F(z)$ and $F(z) - \Delta F(z)$, as shown in Fig. 7(c). As a result, the sequence that drives the DAC has a z -transform given by

$$V_{f,mismatch}(z) = \underbrace{(V_{e2}(z) + V_{o2}(z)) F(z)}_{\mathcal{Z}(v[n])} + \underbrace{(V_{e2}(z) - V_{o2}(z)) \Delta F(z)}_{\mathcal{Z}(\cos[\pi n]v[n])} \quad (3)$$

The first term in the equation corresponds to the ideal sequence that would have excited the DAC in the absence of mismatch. The second term corresponds to the error due

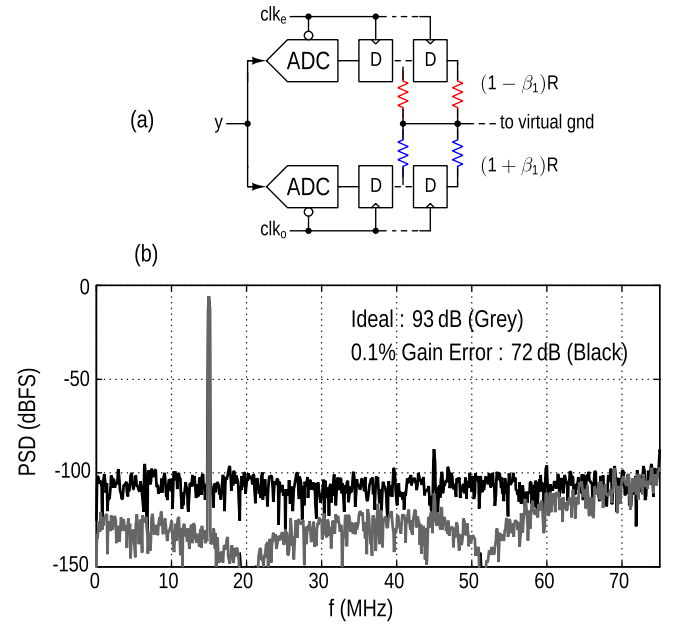


Fig. 8. (a) Modeling gain error in the interleaved sections of the FIR-DAC. (b) Comparison of the in-band PSDs with and without gain mismatch.

to mismatch in the transfer functions of the FIR DACs. The transfer function from the output of the DAC to that of the modulator is unity at low frequencies. Thus, the in-band content of the term $(V_{e2}(z) - V_{o2}(z))\Delta F(z)$ is the cause for concern. From (1), it is seen that $(V_{e2}(z) - V_{o2}(z))$ corresponds to the sequence $\cos[\pi n]v[n]$. In the time domain, therefore, the error corresponds to filtering $\cos[\pi n]v[n]$ with a filter whose transfer function is $\Delta F(z)$. Multiplying $v[n]$ with $\cos[\pi n]$ results in a translation of quantization noise from $f_s/2$ to low frequencies. At in-band frequencies, $\Delta F(z)$ can be approximated by the difference between the DC gains of the FIR filters in the even and odd channels. It is thus seen that the effect of mismatch in the FIR filters is to cause shaped quantization noise from $f_s/2$ to “leak” in-band, by a factor equal to the difference in the DC gain of the FIR filters. The results of Monte-Carlo simulations, where the taps of a time-interleaved FIR DAC have a random mismatch with standard deviation of 1%, confirm this analysis. Fig. 7(d) shows the in-band SNDR for 1000 trials. The degradation in performance due to tap mismatch is about 30 dB. When *only the DC gain* of the interleaved DACs is corrected, the performance greatly improves, as shown in Fig. 7(e). The key point, therefore, is to observe that mismatch in the FIR filter transfer functions is not particularly important, it is mismatch in the DC gain of the time-interleaved FIR-DACs that matters. Fig. 8 compares the in-band PSDs of a CTΔΣM (which uses the equivalent of 8 full-rate taps) with and without gain error. We see that a 0.1% mismatch in the DC gain degrades the in-band quantization noise by about 20 dB, making this by far the most significant non-ideality in a time-interleaved FIR-DAC.

III. ARCHITECTURE AND CIRCUIT DESIGN

Fig. 9 shows the simplified single-ended schematic of the fourth order 1-bit CTΔΣM. The sampling rate is $f_s=6$ GHz, resulting in $OSR = 50$. A cascade of integrators with feed forward and feedback (CIFF-B) loop filter architecture

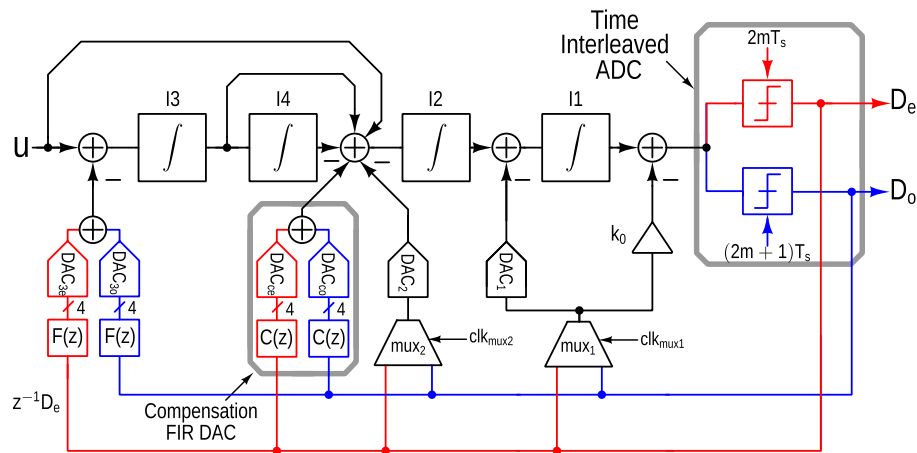


Fig. 9. Block diagram of the fourth order CT $\Delta\Sigma$ M.

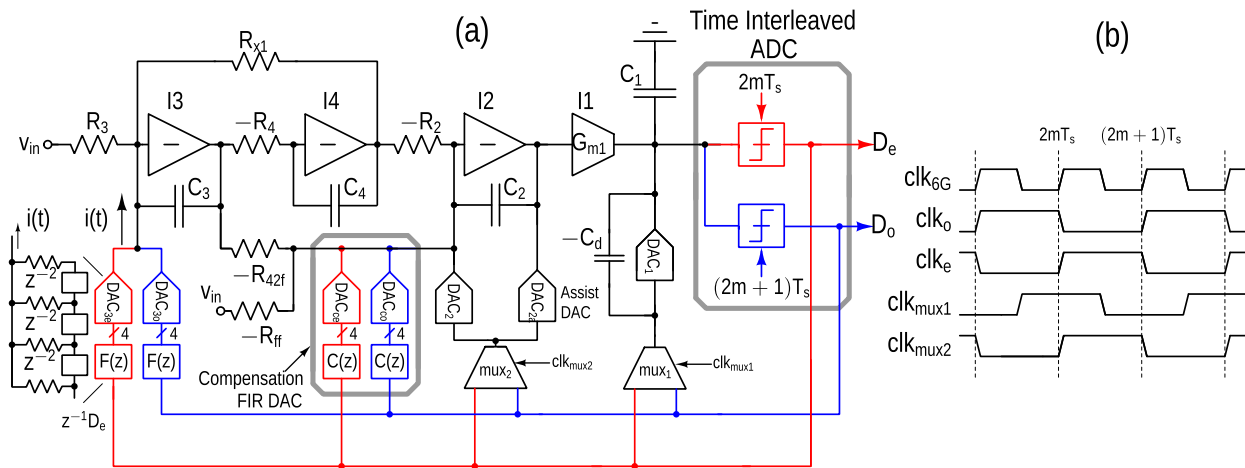


Fig. 10. Simplified single-ended CT $\Delta\Sigma$ M schematic.

is chosen, so that the fast $1/s$ and $1/s^2$ paths around the quantizer are decoupled from the precise $1/s^4$ path. Thanks to feedforward, the gain of the input integrator I3 is large in the signal band: this reduces the noise and distortion of the rest of the loop filter when referred to the input. I3 and I4 are enclosed in a weak feedback loop (not shown in Fig. 9) to realize complex NTF zeros. The loop filter's output is sampled by two time-interleaved comparators. Their outputs D_e and D_o are multiplexed, and drive the direct path (gain k_o) around the quantizer and DAC₁ (which, along with I1 forms the $1/s$ path of the loop filter). The total delay in this path is about three-quarters of a clock cycle. D_e and D_o drive the main TI-FIR DACs, formed by DAC_{3e} and DAC_{3o}. Each of these is a 4-tap FIR DAC, equivalent to 8 full-rate taps. When compared to a full-rate implementation, power dissipation due to the flip-flops in the FIR DAC is reduced by a factor of 2, thanks to half-rate operation. The FIR DAC introduces delay in the loop, rendering the modulator unstable. As shown in [9], the NTF can be restored exactly through the use of a compensation FIR DAC. In our design, the compensation DAC, denoted by DAC_c, is also time-interleaved, so that the flip-flops in DAC₃ can be reused. Due to time-interleaving, the taps of the compensation filter are constrained – however, simulations show that this is inconsequential, and that the NTF can be restored to very nearly the ideally desired one.

Fig. 10(a) shows a simplified single-ended schematic of the modulator. I2, I3 and I4 are realized using active-RC techniques for improved linearity and low noise. I1, which drives the $2\times$ time-interleaved ADC is realized using Gm-C techniques, consisting of G_{m1} and C_1 [10]. This way, the parasitic capacitances of G_{m1} and the quantizer are no longer a burden, as they form part of C_1 . Since the 1b-quantizer output only depends on the sign of its input, variations of C_1 do not compromise robustness. This is in contrast to a multi-bit design, where variations in the quantizer input capacitance, when driven by a transconductor as in [10], would change the quantizer's gain. The direct path around the quantizer is implemented using capacitive division through C_d and C_1 . I2, which is in the $1/s^2$ path of the loop filter, is sped up through the use of opamp assistance via DAC_{2a} [11]. DAC₃ and DAC_c are resistive to achieve low noise. DAC₁ and DAC₂ are current steering DACs which complete the (fast) $1/s$ and $1/s^2$ paths around the quantizer. At high clock rates such as this, the decimation filter will anyway have to be implemented in poly-phase form, so the outputs D_e and D_o need not be recombined into a full-rate bit stream. The time-constants of the active-RC integrators are tuned by digitally-programmable capacitor banks. The $G_{m1} - C_1$ integrator is tuned by digitally programming the transconductor. Due to the feedforward nature of the architecture, a peaking signal trans-

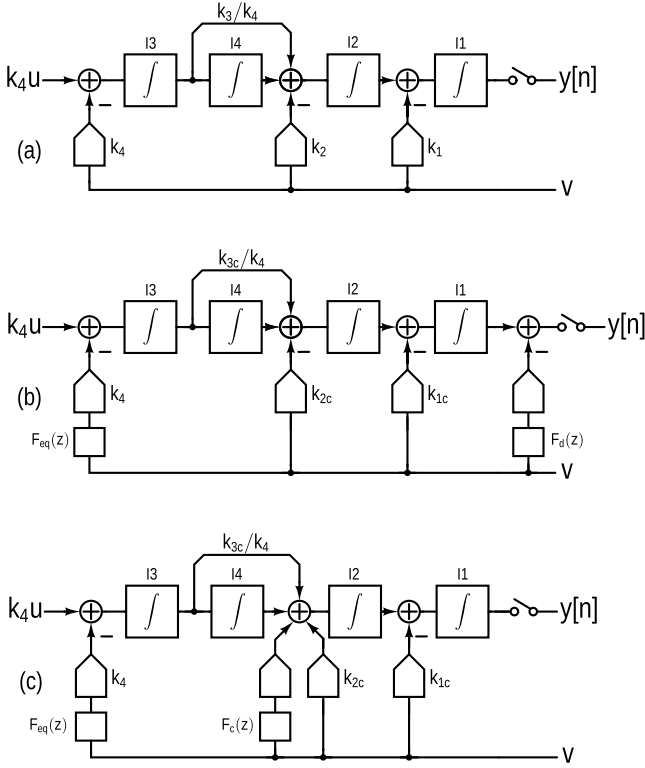


Fig. 12. Evolution of FIR-DAC compensation. (a) Loop filter without FIR DAC. (b) With the main FIR DAC, the NTF can be restored by coefficient tuning and adding a direct-path FIR DAC. (c) Moving the direct path FIR DAC to the $1/s^2$ path.

direct path FIR DAC $F_d(z)$ is then determined so as to make the sampled pulse responses of the loop filter with and without FIR feedback equal for $t < N$.

It turns out that the compensation filter can be moved to input of the $1/s^2$ path, as shown in Fig. 12(c), by appropriately modifying its transfer function to $F_c(z)$. The advantages of doing so are twofold. First, the taps of the compensation filter are smaller. This makes intuitive sense due to the following. The main FIR DAC delays the responses of the $1/s^4$ and $1/s^3$ paths. The deviation between the delay-free and delayed responses is integrated by I_2 and I_1 before being sampled. Thus, “fixing” the error at a later stage, in general, will need filter taps with a larger magnitude. Another benefit of moving the compensation filter to the input of I_2 is that the summing node at the output of the loop filter is avoided, thereby reducing capacitive loading at the quantizer input.

As we discussed earlier, the main FIR DAC is implemented in a time-interleaved fashion. The compensation FIR DAC uses the same flip-flops as the main DAC, thereby constraining successive taps to be identical. Fortunately, simulations show that the error in the compensated NTF due to this is negligible. A full-rate direct path is used to mitigate the finite-bandwidth effect of real OTAs, and the explicit delay introduced into the loop. Component values that account for amplifier bandwidth in the NTF design are arrived at by using the robust design centering procedure described in [12].

D. Calibration

In section II-C, we showed that mismatch between the dc gains of the even and odd FIR DACs causes translation of

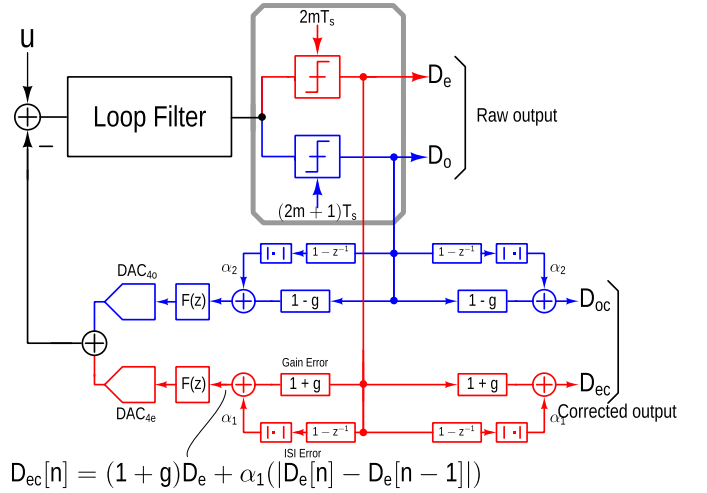


Fig. 13. Non-idealities in the time-interleaved FIR DAC, and the principle of calibration.

shaped quantization noise from $f_s/2$ to in-band frequencies. Another error, arising from the NRZ feedback DACs used, is due to inter symbol interference (ISI). This occurs due to mismatch between the rise and fall times of the feedback waveform. In this work, dc gain-mismatch and ISI were addressed through offline calibration. The simplified model of the modulator incorporating gain and ISI errors is shown in Fig. 13. Gain mismatch between the even and odd DACs can be conceptually thought of as multiplying their input sequences D_e and D_o by $(1 + g)$ and $(1 - g)$ respectively. As shown in [6], using an FIR DAC does not mitigate error due to ISI. There are many ways of modeling ISI error in a single-bit DAC, and one way of doing this is to add a constant to the bit whenever there is a transition. Therefore, ISI in the even DAC output can be modeled by adding a term $\alpha_1(|D_e[n] - D_e[n-1]|)$. The odd DAC's ISI can be similarly modeled. The combined error due to gain-mismatch and ISI in the even DAC can be thought of as *conceptually* exciting an otherwise ideal DAC with a sequence $D_{ec}[n] = (1 + g)D_e[n] + \alpha_1(|D_e[n] - D_e[n-1]|)$. In a similar fashion, the odd DAC can be thought of as being excited by $D_{oc}[n] = (1 - g)D_o[n] + \alpha_2(|D_o[n] - D_o[n-1]|)$.

The loop filter forces the DAC output waveform to equal the input at low frequencies. If we had access to $D_{ec}[n]$ and $D_{oc}[n]$ in Fig. 13, the digital representation of u could be obtained by first forming the full rate sequence $D_{ec}(z^2) + z^{-1}D_{oc}(z^2)$, and decimating it. The gain error g , and ISI error coefficients α_1 and α_2 are unknowns. Ideally, these are all zero. Non-zero values will degrade the in-band SNDR, since a non-zero g translates shaped noise at $f_s/2$ into the signal band, and ISI causes even order distortion and increases in-band noise floor. Thus, given the raw output streams D_e and D_o , these coefficients can be found to be those that maximize the in-band SNDR. The calibration procedure is as follows. g , α_1 and α_2 are initialized to zero. The *corrected* full-rate sequence $D_{ec}(z^2) + z^{-1}D_{oc}(z^2)$ is formed, and its in-band SNDR is computed. Clearly, the SNDR will be a function of g , α_1 and α_2 . The g , α_1 and α_2 that maximize the SNDR are obtained by using the Nelder-Mead simplex optimization algorithm.

TABLE I
PERFORMANCE SUMMARY AND COMPARISON WITH STATE OF THE ART ADCs

	This work	[1]	[2]	[3]	[5]	[6]	[7]	[10]	[14]	[15]
CMOS node	65 nm	65 nm	28 nm	65 nm	28 nm	90 nm	40 nm	45 nm	45 nm	20nm
Area (mm ²)	0.07	0.50	0.90	0.16	0.34	0.12	0.02	0.90	0.49	0.10
Supply Voltage (V)	1.4	N.A.	1.0	1.2	1.2	1.2	N.A.	1.1	1.4	1.2
Sampling Rate(GHz)	6	1.2	3.2	0.9	1.8	3.6	2.4	4	6	2.18
Power(mW)	13.3	54	235	24.7	80.4	15	5.25	260	20	23
Bandwidth(MHz)	60	50	53.3	45	50	36	40	125	60	80
DR(dB)	76	72	88	80	84.5	80	67.8	70	62	73
SNR _{max} (dB)	68.8	71.7	83.1	78.5	76.8	76.4	N.A.	65.5	61.5	70
SNDR _{max} (dB)	67.6	71.5	71.4	75.3	74.9	70.9	66.9	65	60.6	67.5
FoM ₁ (fJ/level)	56.5	176	726	57.7	177	72.7	36.3	715.6	190.3	74.2
FoM ₂ (dB)	164.1	161.2	155.0	167.9	162.8	164.7	165.7	151.8	155.4	162.9

$$\text{FoM}_1 = \frac{\text{Power}}{2 \cdot \text{BW} \cdot 2^{(\text{SNDR} - 1.76)/6.02}}$$

$$\text{FoM}_2 = \text{SNDR}_{\text{max}} + 10 \log\left(\frac{\text{BW}}{\text{Power}}\right)$$

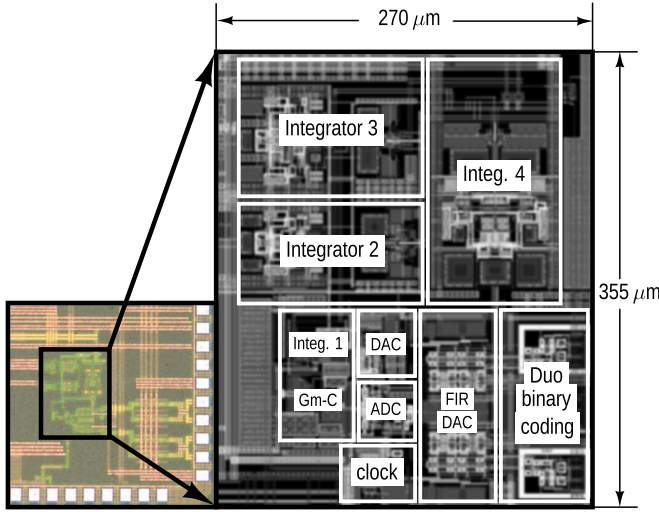


Fig. 14. Chip microphotograph and layout snapshot.

Fortunately, as shown by measurements in the next section, estimating these coefficients at room temperature, and freezing them seem to be adequate, at least for the resolution attempted in this work.

IV. MEASUREMENT RESULTS

The fourth-order CTΔΣM was fabricated in a 65 nm low-leakage process through Europractice. The dies were mounted in a QFN64 package. Fig.14 shows the die photograph and layout of the active area, which measures about 0.07 mm². The two high speed 3 Gb/s single-bit data streams, after duobinary coding, were brought out of the chip using LVDS buffers. The waveforms were captured using a high speed oscilloscope and subsequently processed on a computer. Duobinary coding relaxes the bandwidth requirements of the package, board and the oscilloscope, and reduces the effects of package feedthru on the modulator's performance [13]. The raw bit-streams acquired from the test-chip are calibrated for gain mismatch and ISI errors as described in Sec. III-D.

The SN(D)R, plotted as a function of input amplitude, is shown in Fig.15(a). The peak SNR/SNDR in a 60 MHz bandwidth are 68.8/67.6 dB respectively. The dynamic range is 76 dB. The PSD for a -4.6 dBFS tone at 19.9 MHz is shown in Fig. 15(b). The current drawn by the analog, digital and reference circuitry are 4.05 mA, 3.75 mA and 1.7 mA respectively.

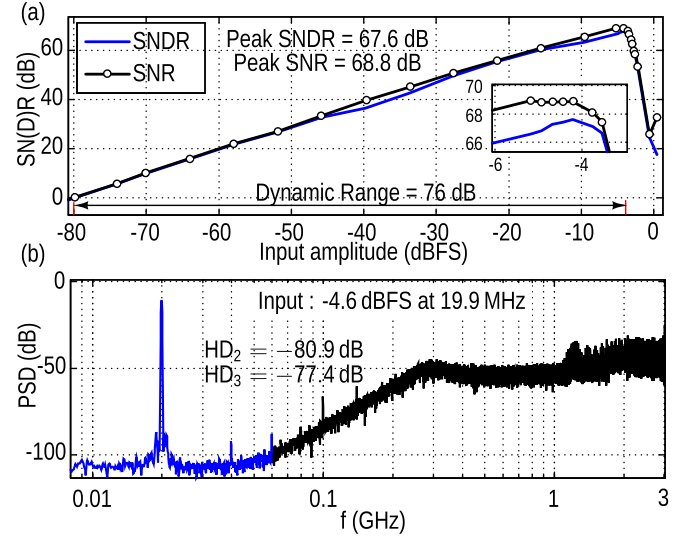


Fig. 15. (a) SN(D)R as a function of input amplitude and (b) PSD for a -4.6 dBFS tone at 19.9 MHz.

The CTΔΣM, clocked at 6 GHz, therefore, consumes 13.3 mW (including power drawn from the references) from the 1.4 V supply. The resulting Walden and Schreier Figures of Merit are 56.5 fJ/level and 172.5 dB respectively.

As described in Sec.III-D, gain mismatch and rise-fall asymmetry of the interleaved sub-DACs are corrected in the digital domain. To determine if the calibration strategy is adequate for the level of resolution attempted in this work, the coefficients (g, α_1, α_2 in Fig. 13) were determined when the CTΔΣM operates at 30°C. Measurements are made by sweeping temperature over the 0–70°C range, but by using coefficients derived at the nominal temperature of 30°C. Fig. 16 shows the peak SNDR, after calibration, remains largely unchanged, indicating that a one point correction is sufficient to account for gain mismatch of the interleaved sub-DACs, and ISI. To assess the relative contributions of gain mismatch and ISI, the peak SNDRs were determined with ISI correction on with mismatch-correction turned off and vice-versa. We see that in this particular design, ISI degrades in-band noise more than gain mismatch in the interleaved DACs. Extrapolating from the simulation results in Fig. 8 would lead us to the conclusion that gain mismatch

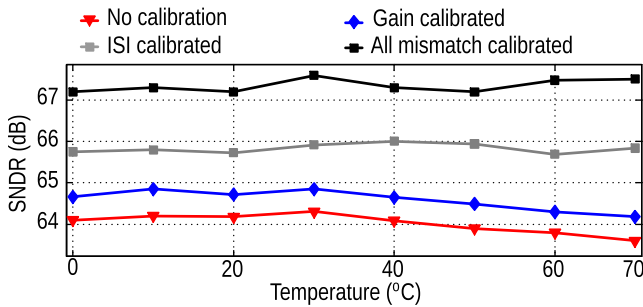


Fig. 16. Peak SNDR variation with temperature under various – the ADC is calibrated at 30°C, and the same coefficients are used across temperature.

is about 0.1%. This seems reasonable, given that the design uses resistive DACs.

Table I compares the performance of this design with that of several high speed $\Delta\Sigma$ modulators presented in the literature. Thanks to time interleaving of quantizer this work achieves figure of merit of 56.53 fJ/1v1 at 6 GHz sampling rate and outperforms several converters targeting similar bandwidths and resolutions, but realized in more advanced processes. Further, barring [7], the active area occupied by our modulator is significantly smaller than that corresponding to many other designs.

V. CONCLUSIONS

We presented design considerations for wide-band single-bit CT $\Delta\Sigma$ M employing time-interleaved FIR feedback. We showed that time-interleaving not only greatly reduces in-band noise due to data-dependent jitter, but also reduces dynamic power dissipation of the FIR DAC by a factor of 2. Like in all interleaved systems, offset, gain and timing mismatches between the time-interleaved sections should be expected to degrade performance. In our case, we showed that the dominant mechanism of degradation of in-band SNDR is the mismatch in the dc gains of the FIR DACs. The efficacy of the architecture and circuit techniques, were borne out from measurements of a single-bit CT $\Delta\Sigma$ M that uses a $2\times$ TI-FIR DAC designed in 65 nm low-leakage CMOS process. The modulator achieves 67.6/68.8/76 dB SNDR/SNR/DR in a 60 MHz bandwidth. Operating at 6 GHz, it occupies only 0.07 mm² and consumes 13.3 mW, resulting a Walden FoM of 56.5 fJ/1v1.

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